

ROSHAN SHUKLA

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer specializing in distributed systems, microservices architecture, and low-latency backend services using Java, Spring Boot, Python, Redis, Docker, and AWS. Experienced in designing scalable APIs, asynchronous processing pipelines, caching systems, and event-driven architectures. Delivered measurable improvements in latency, throughput, reliability, and system performance across production applications.

TECHNICAL SKILLS

Languages: Java, Python, SQL, C, C++

Backend & Cloud: Spring Boot, Microservices, REST APIs, AWS, Docker, CI/CD, GitHub Actions

Systems & Core: Data Structures & Algorithms, System Design, Distributed Systems, Distributed Caching, Concurrent Programming, Multithreading, Design Patterns, Object-Oriented Design

Databases: PostgreSQL, MongoDB, MySQL, Redis

EDUCATION

B.Tech – Computer Science & Engineering (AI/ML)

Nov 2022 – Jun 2026

Dr. A.P.J. Abdul Kalam Technical University (AKTU), Uttar Pradesh

CGPA: 8.0 / 10

EXPERIENCE

Founding Software Engineer *Symptomwise Pvt. Ltd*

Oct 2025 – Jan 2026

- Architected a multi-tenant L1/L2 semantic caching system for LLM inference pipelines using **Redis**, achieving a **61% cache hit rate** and reducing p99 latency from **1.2s** to **213ms**.
- Designed an asynchronous request path to decouple GPU inference from cache hits, reducing contention and maintaining low-latency responses under concurrent workloads.
- Implemented semaphore-based load shedding and circuit breaker mechanisms, maintaining **99.9% availability** during simulated **3x** traffic bursts.

Software Engineering Intern *GanakGyan Technologies*

Jun 2025 – Sep 2025

- Engineered high-performance RESTful APIs using **Java Spring Boot** for a geospatial engine, optimizing **MongoDB** compound indexes to reduce average query execution time by **35%**.
- Conducted load testing and root cause analysis, identifying and resolving thread-pool starvation under peak traffic; automated deployments using **GitHub Actions**, reducing deployment errors by **20%**.

Software Engineering Intern *Karyanest LLP*

Feb 2025 – Apr 2025

- Refactored a legacy **Java Spring Boot** monolith into decoupled microservices, improving fault isolation and enabling independent deployments.
- Built an asynchronous event-driven ingestion pipeline using **Spring Boot** and Redis message queues, increasing system throughput by **250%**.
- Implemented centralized logging, monitoring, and observability tooling, reducing production Mean Time to Detection (MTTD) by over **40%**.

PROJECTS

IntervYou – Real-Time Collaborative Interview Platform

[Live](#) | [GitHub](#)

Next.js • Node.js • Redis Pub/Sub • WebRTC • Docker • Judge0

- Built a real-time synchronization engine using **Redis Pub/Sub** and **Yjs CRDTs** for collaborative code editing, maintaining **110ms p50 latency** across **50+** concurrent users.
- Implemented CRDT-based state synchronization using **Yjs**, ensuring **0% message loss** and sub-**1s** recovery for reconnecting clients.
- Sandboxed multi-language code execution using **Docker + Judge0** across **20+** languages while maintaining host security isolation.

Automated Reddit to YouTube Content Pipeline

[GitHub](#)

Python • FFmpeg • Whisper • Bark TTS • Telegram Bot API

- Engineered a Python automation pipeline using **FFmpeg** and **Whisper** to automate video generation workflows, reducing manual production time by **90%**.
- Built a SQLite-backed deduplication engine to track processed assets, ensuring **100% reliability** while eliminating redundant compute costs.

ACHIEVEMENTS

- LeetCode:** Peak Contest Rating **1736 (Top 11% globally)**, Solved **500+ DSA problems**.
- Leadership:** Elected IEEE Student Branch Secretary; drove technical roadmaps, managed resource scheduling, and spearheaded events for 15+ peer engineers.
- Competitions:** 2x Runner Up at TechYuva (8.0 & 10.0 scale) competing cross-functionally against 50+ engineering cohorts.
- Certifications:** Microsoft Generative AI, Machine Learning (IIT BHU), AWS ML Foundations.